

**AIRCRAFT INTERIOR DESIGN LAB****IV Semester**

| Course Code   | Category   | Hours / Week |          |          | Credits  | Maximum Marks |            |              |
|---------------|------------|--------------|----------|----------|----------|---------------|------------|--------------|
| <b>A6AE12</b> | <b>PCC</b> | <b>L</b>     | <b>T</b> | <b>P</b> | <b>C</b> | <b>CIA</b>    | <b>SEE</b> | <b>Total</b> |
|               |            | -            | -        | 3        | 1.5      | 40            | 60         | 100          |

**COURSE OBJECTIVES:**

1. To design various Aircraft interior components in lab.
2. To design various individual components, sub-assemblies and main assemblies in lab.
3. Discuss the importance of design process and studying the different phases of designing process involved in the design.
4. Understand the design of aircraft concepts.
5. To design various aircraft components by using Catia software

**LIST OF EXPERIMENTS**

- 1 Aircraft Seating Design Arrangements (General Class & Business Class) (2D)
- 2 Aircraft Single Aisle Design(3D & 2D)
- 3 Aircraft Twin Aisle Design(3D & 2D)
- 4 Aircraft Seat Design (3D- Business Class)
- 5 Aircraft Seat Design (3D- First Class)
- 6 Design of Laptop Tray Sizing for Seating in Civil Aircrafts.
- 7 Design of Window for Civil Aircraft
- 8 Design of Wind Shield for Fighter Aircraft(Select any one airplane)
- 9 Design of Wind shield for Passenger Aircraft(Select any one Airplane)
- 10 Design of Arm rest cap height from surface in Civil Aircraft
- 11 Design of Joystick(Spacing) in Fighter Aircraft

**Note:** Ten experiments should be performed.

**Reference Books:**

Basics in Catia V5 with Simulation by Ranjan Chikesh

**COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:**

1. Understand the importance of drawing and design process and phases involved in the design process
2. Design various individual components, sub-assemblies and main assemblies in design lab
3. Design various orthographic and isometric projections in drawing sheets
4. Develop the basic concepts of aircraft interiors
5. Design and develop aircraft interior components